Conventional pansharpening

Launcher: `Toolbox/pansharpening.bat`

Purpose

Pansharpening transfers the finer spatial detail of a single-band panchromatic (PAN) image to a lower-resolution multispectral (MUL) image. It can make small crop-mark patterns easier to see, but it creates a derived image and can introduce spectral or edge artefacts.

Inputs and parameters

Option	What to provide	Behaviour
Input MUL Image	Corrected multiband GeoTIFF	The image is aligned to the PAN grid internally. Use the matching acquisition and footprint.
Input PAN Image	Corrected single-band panchromatic GeoTIFF	Provides the output grid and spatial detail. It should share the MUL image's CRS and acquisition.
Output Folder	Folder, or blank	Blank writes beside the MUL image. A missing entered folder is created only at its final level.
Vector layer	Optional clip polygon	Crops the final pansharpened results (including the WV3 NIR2 product) and assigns NoData `−9999`. The unclipped temporary results are removed. Inputs that were already clipped during atmospheric correction are also supported: their NoData is kept out of the calculation.

Option	What to provide	Behaviour
Sensor	`WV3` or `LEGION`	Used by weighted Brovey. Default: `WV3`. Sensor choice does not affect Bayesian processing.
Algorithm	`Bayesian` or `wBrovey`	Selects the fusion method. Default: `Bayesian`.
MUL resampling	`Nearest`, `Bilinear` or `Cubic`	How the MUL image is resampled onto the PAN grid before the fusion. Applies to every algorithm, including the WV3 NIR2 LMVM step. Default: `Bilinear`. See Choosing a MUL resampling method.
lambda	Floating-point value from 0 to 1	Bayesian regularization/weighting parameter. Default: `0.995`. The field is validated even when `wBrovey` is selected.
Output compression	`Lossless` or `Near-lossless`	Lossless (default) is bit-exact. Near-lossless (LERC) allows an error of at most 5e-5 per pixel and only applies to Float32 outputs; when both inputs hold integers, the output is an integer and always lossless. Keep the output lossless if it feeds PCA, spectral indices or MAE: see Output compression.

Algorithms

Bayesian uses Orfeo ToolBox's Bayesian method. The method uses whole-image statistics, so on inputs with NoData the values can differ by a fraction of a percent from a run on unclipped inputs. Higher lambda values impose a stronger PAN-driven constraint. The default is intended for reflectance

scaled to integer-like values; with unscaled 0–1 reflectance, it can overwhelm the spectral signal and produce nearly identical bands or extreme values.

wBrovey injects PAN detail using sensor-specific spectral weights. WV3 uses seven weighted bands, then pansharpens NIR2 separately with local mean/variance matching and creates a VRT stack. The current LEGION sensor has no Brovey weights configured, so `wBrovey` is not operational for LEGION; use Bayesian instead.

Choosing a MUL resampling method

The MUL pixels are larger than the PAN pixels (for WV3, 1.24 m against 0.31 m, so each MUL pixel covers 4×4 PAN pixels). Before the fusion, the MUL image is resampled onto the PAN grid. The fusion adds PAN detail on top of this resampled image, so how it is resampled stays visible in the output.

Method	What it does	Strengths	Drawbacks
`Nearest`	Copies each MUL pixel into all the PAN pixels it covers.	Every pixel keeps an original MUL value; no new spectra are created.	Leaves square blocks the size of a MUL pixel (4×4 output pixels for WV3), most visible in smooth areas and with Bayesian. These are artefacts, not ground features.
`Bilinear` (default)	Weighted average of the 4 nearest MUL pixels.	Smooth, no blocks. Never goes outside the range of the neighbouring values.	Slightly softens MUL edges. Mixes neighbouring spectra at boundaries (for example, a field edge gets intermediate values).
`Cubic`	Cubic convolution over the 16 nearest MUL pixels.	Smooth and sharper than bilinear. Orfeo ToolBox's own default for this step.	Can overshoot at strong edges: thin bright or dark halos, and values slightly below 0 next to very dark pixels.

- Use `Bilinear` for general visual interpretation, and for spectral indices or PCA computed on the pansharpened image.
- Use `Cubic` when you want the sharpest result and can check strong edges for halos.
- Use `Nearest` only when every output pixel must hold an original MUL value (for example, to compare against the unsharpened data pixel by pixel). Do not interpret its square patterns as archaeological features.

In all three methods, MUL pixels with NoData are left out of the interpolation, so they are never blended into valid pixels.

Outputs

- Bayesian: `<MUL-name>_bayes.tif`
- Weighted Brovey: `<MUL-name>_brovey.tif`
- WV3 weighted Brovey also creates an NIR2 LMVM product and a `HIGHRES_stack.vrt` that combines the products.

Pixels where an input has NoData are NoData in the outputs (`-9999`, or `0` for unsigned integer outputs).

NIR2 edge margin

The NIR2 LMVM product is also NoData within 3 pixels of an input NoData pixel. LMVM computes each output pixel from the mean and variance of the PAN and NIR2 values in a window of 3 pixels around it. Next to a NoData area, that window includes pixels that hold no real data, so the local statistics are wrong and single pixels can be off by 3–5 % of their value. Weighted Brovey works pixel by pixel and has no such window, so bands 1–7 keep their full extent.

The margin only appears when the inputs already contain NoData, for example when the images were clipped during atmospheric correction. Band 8 of `HIGHRES_stack.vrt` then ends 3 pixels (about 1 m for WV3) inside the edge of bands 1–7; use band 8 as the mask when all eight bands are needed. Inputs without NoData are not affected, and neither is a clip made with this tool's **Vector layer** option, because that clip runs after the fusion and gives all eight bands the same extent.

Main outputs are Cloud Optimized GeoTIFFs: DEFLATE (lossless), or LERC (near-lossless, Float32 only). Intermediate alignment and calculation rasters are deleted after successful processing.

Recommended settings and checks

- For DOS3 outputs scaled by `10000`, begin with Bayesian lambda `0.995`.

- For reflectance in the 0-1 range, begin near `0.1`, then compare results rather than assuming one value is universally correct.
- Check that output bands have plausible and distinct min/max, mean, and standard deviation values. Nearly identical statistics in every band indicate spectral collapse.
- Regular square patterns that line up with the MUL pixel grid come from `Nearest` resampling; rerun with `Bilinear` or `Cubic`.
- Compare sharp linear features with the original PAN. Doubled edges, halos, and shifts may indicate input misregistration rather than archaeological structure.
- Keep the unsharpened multispectral data for quantitative spectral analysis; pansharpened pixels are modelled values.